

Titanic Exploratory Data Analysis

Business Insights Report — Project 2

This report presents the exploratory data analysis performed on the cleaned Titanic dataset, including feature engineering, key visualizations, statistical analysis, and the business insights and recommendations derived from the findings.

Data Understanding

This dataset holds the data of the passengers aboard the RMS Titanic in April, 1912. It contains a total of 891 rows and 11 columns containing their personal information, their tickets and destination, the number of family members they had, and whether they survived or not.

The features are a mix of numerical columns (Age, Fare, Siblings_and_Spouses, Parents_and_Children), categorical columns (Sex, Ticket_Class, Port_of_Embarkation), and identifier/text columns (Passenger_ID, Name, Ticket). The target column, Survived, is binary (0 = did not survive, 1 = survived). Since this is the cleaned dataset from Project 1, there are no missing values or duplicate rows to handle.

Overall, the dataset gives a solid mix of demographic and travel-related information that can be used to explore which factors (such as gender, class, age, and fare) were associated with a higher chance of survival.

Feature Engineering

Feature	How It Was Created	Why It Was Created
Family_Size	Siblings_and_Spouses + Parents_and_Children + 1	Combines two split columns into one number, making it easy to see a passenger's total family size at a glance.
Is_Alone	'Yes' if Family_Size = 1, else 'No'	Allows direct comparison of survival rates between passengers travelling alone and with family.
Age_Group	Child, Teenager, Adult, Senior Citizen bins	Converts continuous Age into life-stage categories to compare survival patterns and identify vulnerable groups.
Fare_Category	Low, Medium, High fare bins	Makes it easier to examine whether passengers paying higher fares had greater chances of survival.

Exploratory Data Analysis

Q1. What percentage of passengers survived?

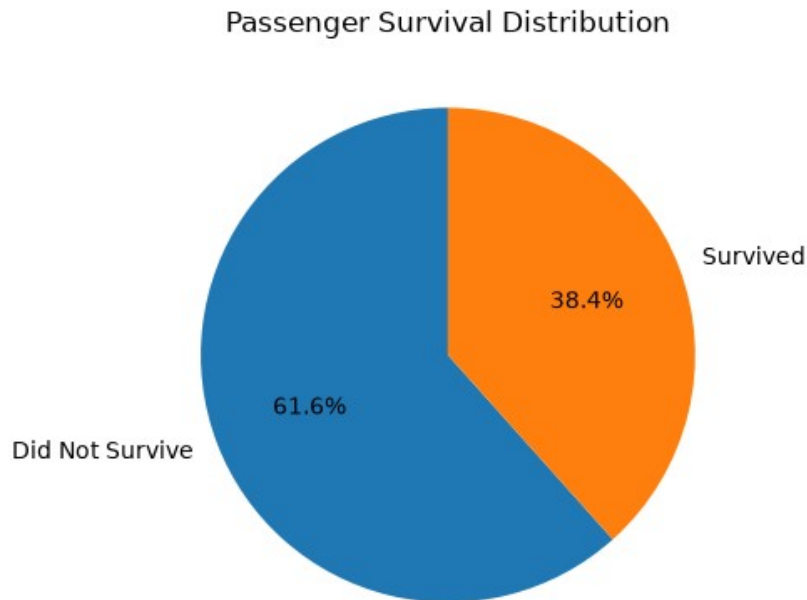


Figure 1: Passenger survival distribution

Observation: Only 38.38% of passengers survived, while the majority (61.62%) did not. This shows that survival was not the outcome for most passengers.

Q2. Which gender had the highest survival rate?

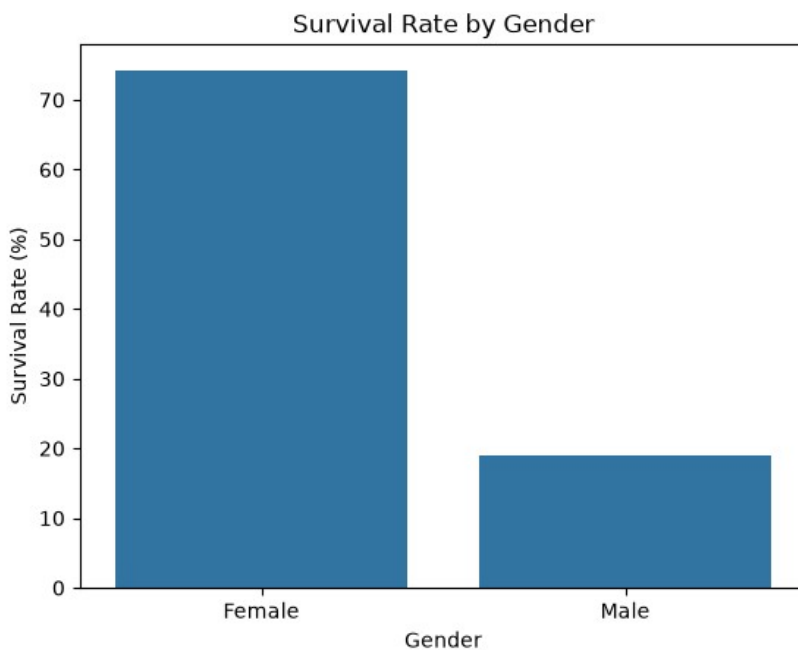


Figure 2: Survival rate by gender

Observation: Females had a dramatically higher survival rate (74.20%) than males (18.89%). This shows a strong difference in survival rates between male and female passengers.

Q3. Which passenger class had the highest survival rate?

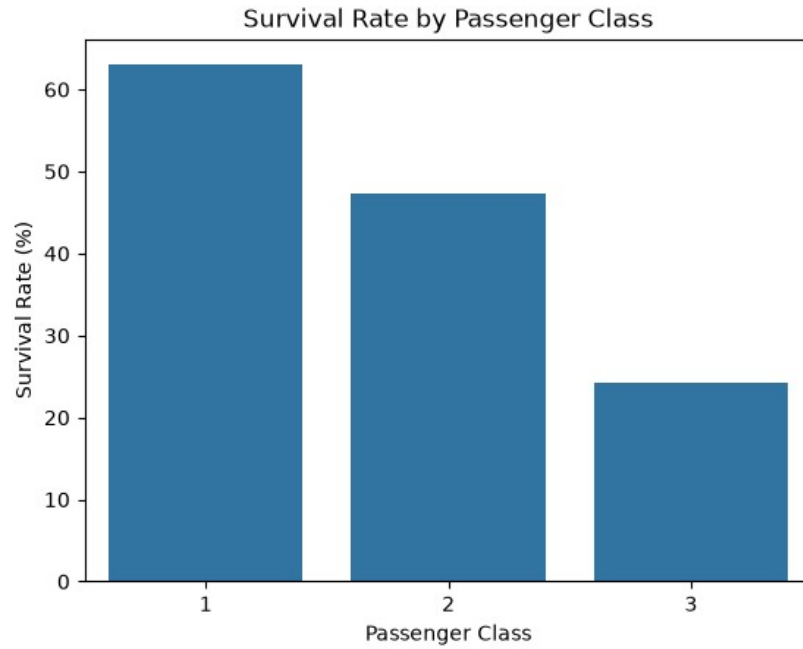


Figure 3: Survival rate by passenger class

Observation: First class passengers had the highest survival rate (62.96%), followed by second class (47.28%), while third class passengers had the lowest by far (24.24%). This shows a clear difference in survival rates across passenger classes.

Q4. Does age affect survival?

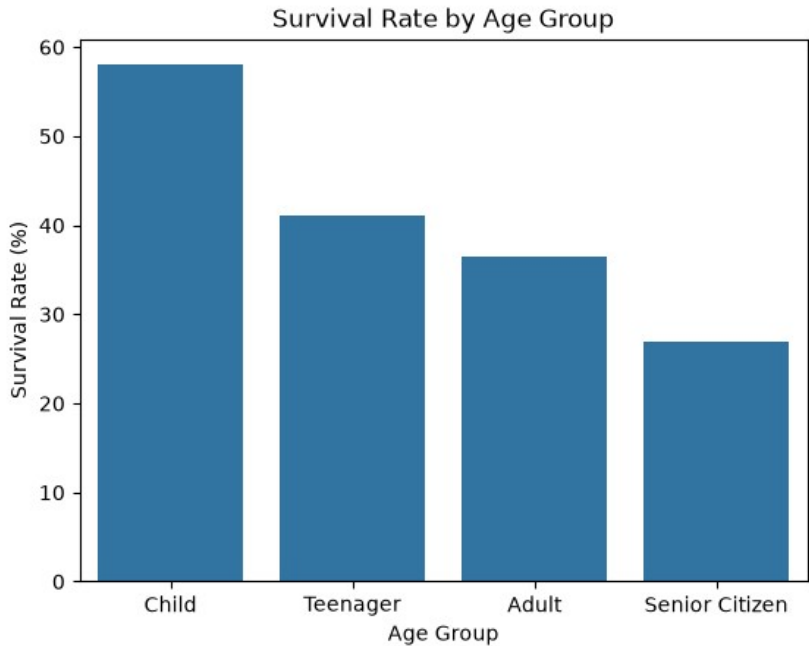


Figure 4: Survival rate by age group

Observation: Children had the highest survival rate (57.97%), followed by Teenagers (41.05%), Adults (36.52%), and Senior Citizens had the lowest (26.92%). This suggests that age was associated with survival, with younger passengers generally having higher survival rates.

Q5. Does fare influence survival?

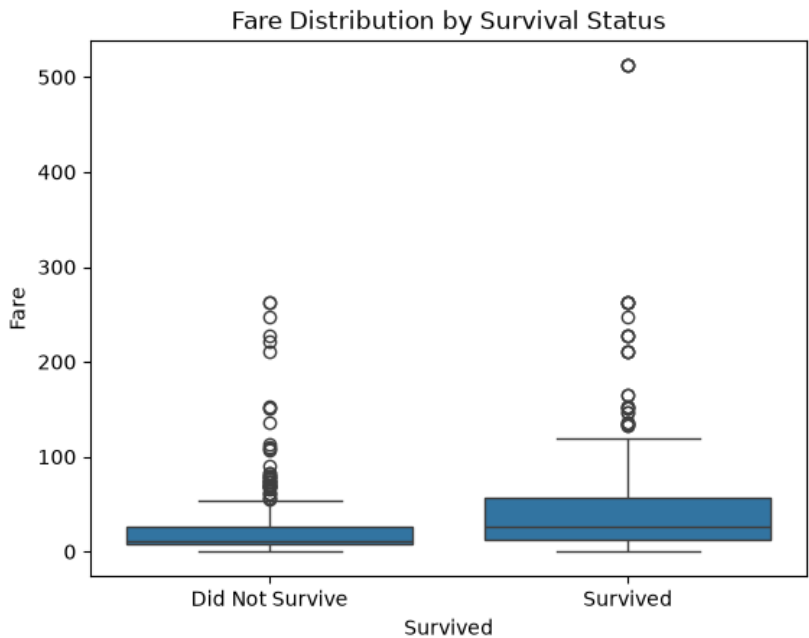


Figure 5: Fare distribution by survival status

Observation: Higher fares were associated with higher survival rates. Passengers in the "High" fare category survived at 76.29%, compared to 45.57% for "Medium" fare and only 28.73% for "Low" fare. This relationship may partly reflect the relationship between fare and passenger class, since first-class passengers generally paid higher fares.

Q6. Did passengers travelling alone survive more often?

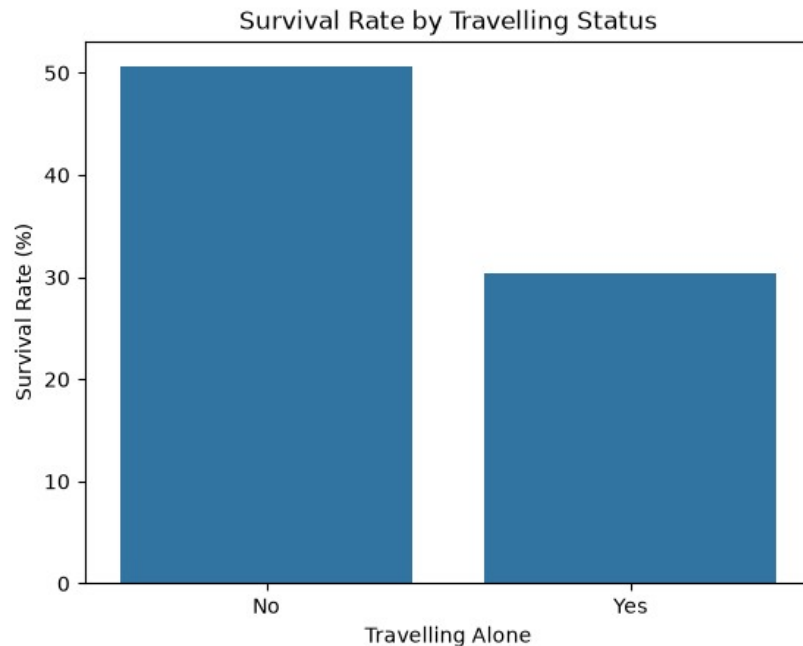


Figure 6: Survival rate by travelling status

Observation: Passengers travelling with family had a higher survival rate (50.56%) than those travelling alone (30.35%), showing an association between travelling status and survival.

Q7. Which embarkation port had the highest survival rate?

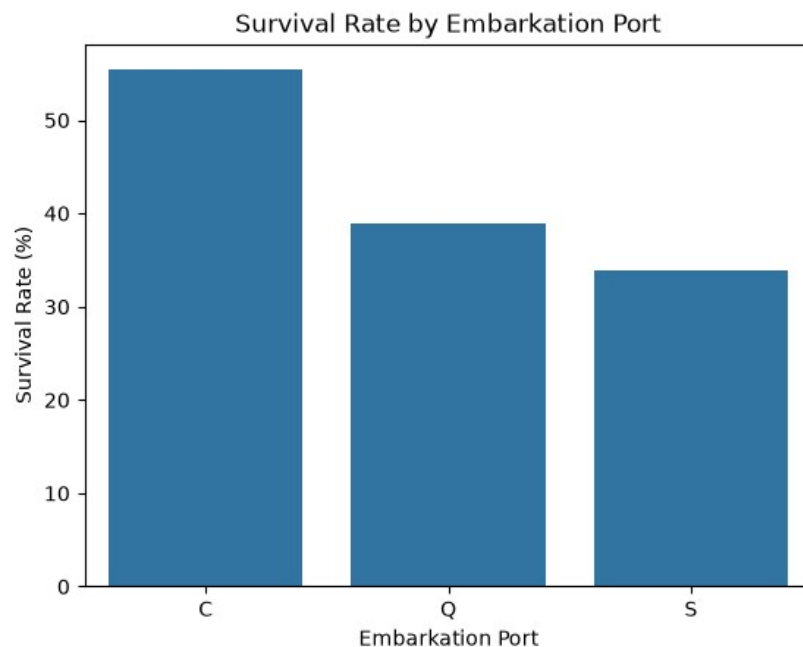


Figure 7: Survival rate by embarkation port

Observation: Passengers who boarded at Cherbourg (C) had the highest survival rate (55.36%), followed by Queenstown (Q) at 38.96%, while Southampton (S) had the lowest at 33.90%. This is likely connected to class distribution, since a larger proportion of Cherbourg passengers travelled in first class.

Q8. What is the relationship between passenger class and fare?

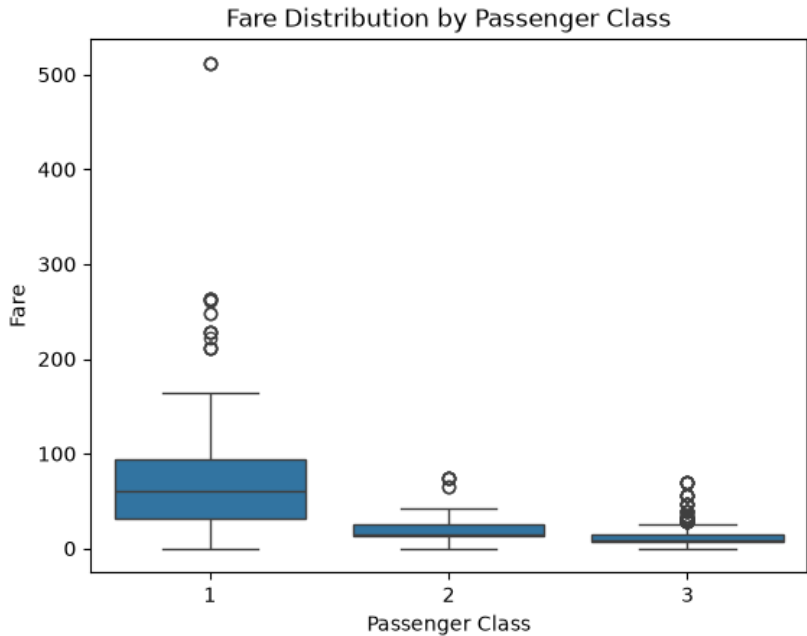


Figure 8: Fare distribution by passenger class

Observation: There is a strong relationship between passenger class and fare. First class passengers paid noticeably higher fares (with several high outliers), followed by second class, with third class fares being the lowest and most tightly clustered.

Q9. Correlation Heatmap

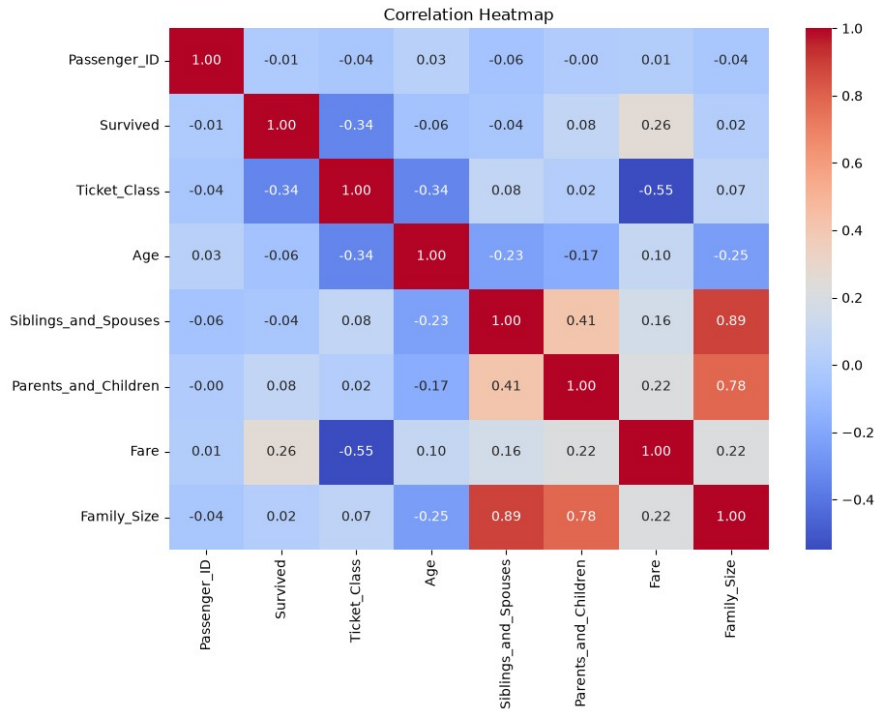


Figure 9: Correlation heatmap of numerical features

Explanation: The strongest positive correlation with Survived is Fare (+0.26), meaning passengers who paid more tended to survive more often. The strongest negative correlation with Survived is Ticket_Class (-0.34), meaning passengers in a lower class tended to survive less often. There is also a very strong positive correlation between Siblings_and_Spouses and Family_Size (+0.89) and between Parents_and_Children and Family_Size (+0.78), which is expected since Family_Size is

built from those two columns. There is also a fairly strong negative correlation between Ticket_Class and Fare (-0.55), reinforcing that higher class paid higher fares.

Q10. Additional Analysis — Family Size vs Survival

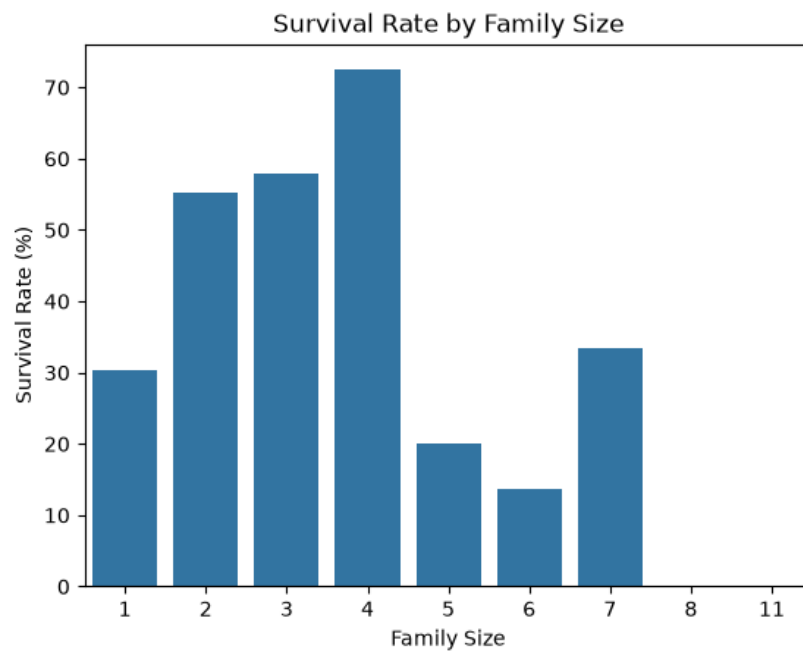


Figure 10: Survival rate by family size

Finding: Survival rate rises with family size up to a point, then drops off. Passengers with a family size of 2 to 4 had the highest survival rates (55–72%), while solo travellers (family size 1) survived at only 30.35%, and passengers in very large families (5 or more) survived at much lower rates, some as low as 0%. This shows that survival rates varied by family size, with passengers in smaller families generally having higher survival rates than solo travellers and those in very large families.

Statistical Analysis

Column	Mean	Median	Mode	Std Dev
Age	29.36	28.00	28.00	13.02
Fare	32.20	14.45	8.05	49.69
Family_Size	1.90	1.00	1.00	1.61

Age: The mean (29.36) is close to the median and mode (both 28), showing only a slight right skew from a smaller number of older passengers. The standard deviation (13.02) shows a fairly wide spread of ages aboard.

Fare: The mean (32.20) is much higher than the median (14.45) and mode (8.05), showing a strong right skew. Most passengers paid relatively low fares, while a small number of first class passengers paid very high fares that pull the average up. The large standard deviation (49.69) confirms this wide spread.

Family_Size: The mean (1.90) is close to the median and mode (both 1), showing that most passengers travelled alone or with only one other family member, while a smaller number of large families pull the mean slightly higher. The standard deviation (1.61) is relatively small, showing family size doesn't vary as wildly as fare.

Dashboard

The dashboard below brings together the most important visualizations from the analysis in one place: survival, gender, class, age, and fare distributions, along with the correlation heatmap.

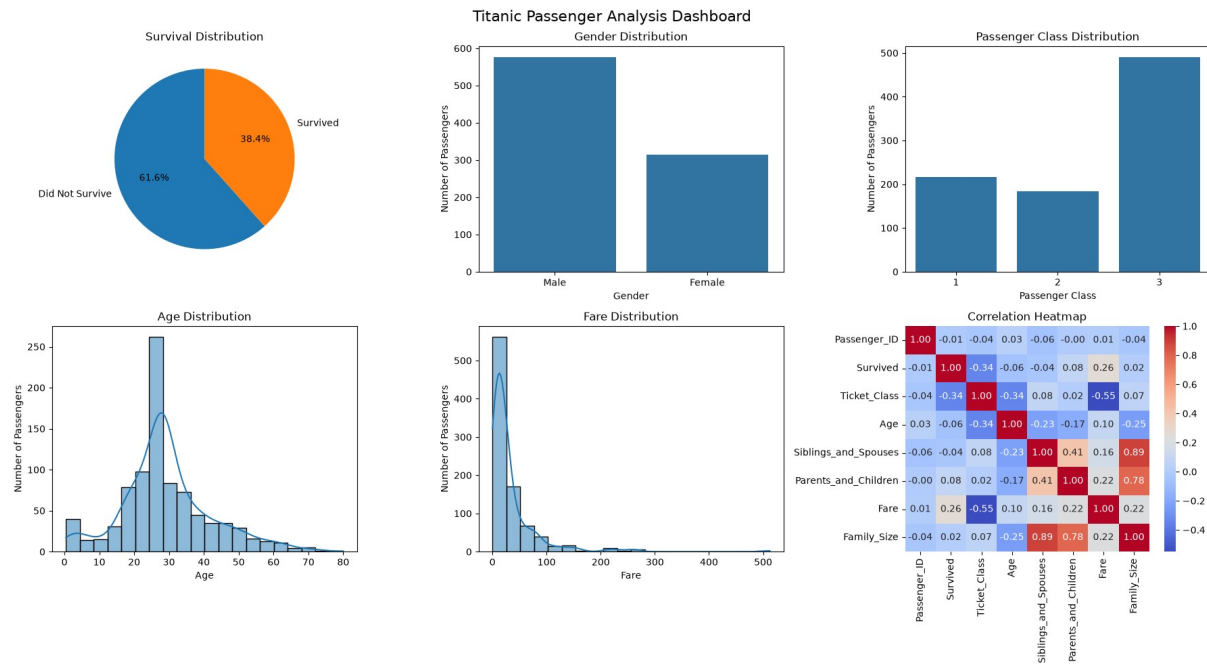


Figure 11: Summary dashboard of key distributions and correlations

Business Insights

1. Most passengers did not survive. Only 38.38% of passengers survived, while 61.62% did not.
2. Female passengers had a much higher survival rate than males. Their survival rates were 74.20% and 18.89%, respectively.
3. Survival was higher among first-class passengers. First-class passengers had a 62.96% survival rate, compared with 24.24% for third-class passengers.
4. Younger passengers generally had higher survival rates. Children had the highest survival rate at 57.97%, while senior citizens had the lowest at 26.92%.
5. Higher-fare passengers had higher survival rates. High-fare passengers had a 76.29% survival rate, compared with 28.73% for low-fare passengers.
6. Passengers travelling with family had higher survival rates. Their survival rate was 50.56%, compared with 30.35% for passengers travelling alone.
7. Survival rates differed by embarkation port. Passengers from Cherbourg had the highest survival rate at 55.36%, while Southampton had the lowest at 33.90%.
8. Passenger class and fare were strongly related. Higher-class passengers generally paid higher fares, showing a clear relationship between the two variables.
9. Family size was related to survival. Passengers in smaller families generally had higher survival rates, while very large families had lower survival rates.
10. Fare and passenger class showed the strongest relationships with survival. Fare had a positive correlation with survival (+0.26), while passenger class had a negative correlation (-0.34).

Business Recommendations

1. **Improve emergency access for third-class passengers.** Since third-class passengers had the lowest survival rate, their evacuation routes and access to lifeboats should be improved.
2. **Provide clear evacuation support for vulnerable passengers.** Emergency procedures should ensure that passengers who need additional assistance can evacuate safely.
3. **Provide more support for families during emergencies.** Since passengers travelling with family had higher survival rates, staff should help families stay together during evacuations.
4. **Provide equal safety resources across all passenger classes.** Since survival differed greatly by passenger class and fare, safety equipment and evacuation access should be available equally to all passengers.
5. **Provide extra assistance to passengers travelling alone.** Since solo passengers had a lower survival rate, staff could give them additional guidance and support during emergencies.